The relationship between a non-Western migration background and education level

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2026-06-22

Set-up your environment

```
## Loading required package: tidyverse

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
## Loading required package: rmarkdown
##
## Loading required package: yaml
##
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
##
##
## Attaching package: 'gridExtra'
##
##
## The following object is masked from 'package:dplyr':
##
##     combine
##
##
##
## Attaching package: 'renv'
##
##
## The following object is masked from 'package:rmarkdown':
##
##     run
##
##
## The following object is masked from 'package:purrr':
```

```
##
##     modify
##
##
## The following objects are masked from 'package:stats':
##
##     embed, update
##
##
## The following objects are masked from 'package:utils':
##
##     history, upgrade
##
##
## The following objects are masked from 'package:base':
##
##     autoload, load, remove, use
```

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Part 1 - Identify a Social Problem

1.1 Describe the Social Problem

A study by research agency SCALIQ involving 6409 secondary school students, suggests that students with Arabic first and/or last names are more likely to be placed at a lower educational level than equally intelligent student with European sounding names. About half of the students with Arabic names were placed half a school level lower. The researchers say that this may be a sign of inequality in education. This is a problem because it can hurt students' education, career opportunities and motivation. However, the study only looks at intelligence and names. It did not look at other factors such as family income, home support or cultural background. The experts say that being placed in a lower school level that is too low can hurt students' education, confidence/motivation and career opportunities.

In another study by KIS they researched whether students with a migration background in the Netherlands receive a lower educational recommendation than they deserve for their abilities. Based on quantitative analyses and interviews with teachers, students and parents, the researchers found just a little evidence of discrimination based on ethnicity. However, factors such as Dutch language proficiency, home environment, classroom behavior and of course the perceptions of the teachers can influence the recommendations. This may negatively impact children with a different ethnic background.

These studies demonstrate that, to some extent, non-Western migrants in the Netherlands seem to obtain a lower level of education than the Dutch. While causal relationships are harder to determine, this gap in education is a social problem as it fuels income inequality, inequality in general, and disadvantages people with a migration background. The current politics and the widespread attention in the Netherlands for migrants, especially non-Western migrants, make this a relevant problem.

Part 2 - Data Sourcing

2.1 Load in the data

2.2 Provide a short summary of the dataset(s)

Summary dataset on education level and migration background

In the first dataset, which contains data on different education levels and migration background, a group consisting of both men and women was researched. All participants were 15 years or older. The dataset contains observations from 2003 to 2020 and contains data about the migration background, age, and level of education of participants for each year. It was sourced from the CBS (2021).

```
summary(edu_mig_dataset)
```

```
##        ID              Geslacht          Leeftijd     Migratieachtergrond
##  Min.   :   190   Length   :12474   Min.   :20150   Length   :12474
##  1st Qu.: 41410   N.unique :    1   1st Qu.:52052   N.unique :   11
##  Median : 83003   N.blank  :    0   Median :53500   N.blank  :    0
##  Mean   : 82923   Min.nchar:    7   Mean   :46186   Min.nchar:    7
##  3rd Qu.:124316   Max.nchar:    7   3rd Qu.:53800   Max.nchar:    7
##  Max.   :165536                     Max.   :53925
##  HoogstBehaaldOnderwijsniveau      Perioden      Bevolking_1
##  Min.   :2018710              Length   :12474   Min.   :   0.0
##  1st Qu.:2018720              N.unique :   18   1st Qu.:   2.0
##  Median :2018770              N.blank  :    0   Median :  16.0
##  Mean   :2018764              Min.nchar:    8   Mean   : 130.7
##  3rd Qu.:2018800              Max.nchar:    8   3rd Qu.:  77.0
##  Max.   :2018810                                Max.   :3469.0
```

Summary dataset on migration background percentage per region

This dataset contains information of the percentage distribution of people with different backgrounds living in each province for 2026. It was sourced from AlleCijfers (2026).

```
##    Provincienaam % Herkomst Nederland % Herkomst buiten Nederland
##  Length   :12    Min.   :0.6100       Min.   :0.1300
##  N.unique :12    1st Qu.:0.6950       1st Qu.:0.1900
##  N.blank  : 0    Median :0.7650       Median :0.2350
##  Min.nchar: 7    Mean   :0.7475       Mean   :0.2525
##  Max.nchar:13    3rd Qu.:0.8100       3rd Qu.:0.3050
##                  Max.   :0.8700       Max.   :0.3900
##  % Herkomst Europa % Herkomst buiten Europa
##  Min.   :0.05000   Min.   :0.0800
##  1st Qu.:0.06750   1st Qu.:0.1150
##  Median :0.08500   Median :0.1300
##  Mean   :0.08917   Mean   :0.1633
##  3rd Qu.:0.11250   3rd Qu.:0.2175
##  Max.   :0.14000   Max.   :0.3000
##  % Geboren in Nederland met herkomst Nederland
##  Min.   :0.6100
##  1st Qu.:0.6950
```

```
##  Median :0.7650
##  Mean   :0.7475
##  3rd Qu.:0.8100
##  Max.   :0.8700
##  % Geboren in Nederland met herkomst Europa
##  Min.   :0.0200
##  1st Qu.:0.0300
##  Median :0.0300
##  Mean   :0.0325
##  3rd Qu.:0.0325
##  Max.   :0.0600
##  % Geboren in Nederland met herkomst buiten Nederland
##  Min.   :0.050
##  1st Qu.:0.080
##  Median :0.095
##  Mean   :0.105
##  3rd Qu.:0.135
##  Max.   :0.170
##  % Geboren in Nederland met herkomst buiten Europa % Geboren in Nederland
##  Min.   :0.0300                                    Min.   :0.7700
##  1st Qu.:0.0475                                    1st Qu.:0.8325
##  Median :0.0550                                    Median :0.8500
##  Mean   :0.0700                                    Mean   :0.8517
##  3rd Qu.:0.1025                                    3rd Qu.:0.8900
##  Max.   :0.1400                                    Max.   :0.9200
##  % Geboren buiten Nederland met herkomst Europa
##  Min.   :0.03000
##  1st Qu.:0.04000
##  Median :0.05500
##  Mean   :0.05583
##  3rd Qu.:0.07250
##  Max.   :0.09000
##  % Geboren buiten Nederland met herkomst buiten Europa
##  Min.   :0.05000
##  1st Qu.:0.06750
##  Median :0.07500
##  Mean   :0.09167
##  3rd Qu.:0.12000
##  Max.   :0.16000
##  % Geboren buiten Nederland
##  Min.   :0.0800
##  1st Qu.:0.1100
##  Median :0.1500
##  Mean   :0.1483
##  3rd Qu.:0.1675
##  Max.   :0.2300
```

Summary dataset on education level per region

This dataset contains information on the participation by people in different levels of education, expressed as a percentage, for each province. It contains data for the years 2023, 2024, and 2025. It was sourced from CBS (2025).

```r
summary(reg_edu_dataset)
```

```
##       Regio's      Perioden
##  Length   :40   Min.   :2023
##  N.unique :14   1st Qu.:2023
##  N.blank  : 0   Median :2024
##  Min.nchar: 9   Mean   :2024
##  Max.nchar:18   3rd Qu.:2025
##                 Max.   :2025
##                 NAs    :1
##  Onderwijs|Naar woonregio|Hoogstbehaald onderwijsniveau|Basisonderwijs, vmbo, mbo1 (%)
##  Min.   :22.30
##  1st Qu.:25.93
##  Median :26.55
##  Mean   :26.18
##  3rd Qu.:27.27
##  Max.   :28.80
##  NAs    :14
##  Onderwijs|Naar woonregio|Hoogstbehaald onderwijsniveau|Havo, vwo, mbo2-4 (%)
##  Min.   :35.50
##  1st Qu.:40.83
##  Median :42.70
##  Mean   :42.51
##  3rd Qu.:44.92
##  Max.   :46.80
##  NAs    :14
##  Onderwijs|Naar woonregio|Hoogstbehaald onderwijsniveau|Hbo, wo (%)
##  Min.   :24.90
##  1st Qu.:27.75
##  Median :31.75
##  Mean   :31.32
##  3rd Qu.:33.10
##  Max.   :42.20
##  NAs    :14
```

```r
inline_code = TRUE
```

2.3 Describe the type of variables included

The first dataset mentioned, edu_mig_dataset, contains demographic and SES information. The demographic information consists of age, sex, and background and the socioeconomic status information consists of education level. The two central variables in this dataset are background and education level, however, the additional information provides possibilities for more thorough analysis. The information has been obtained through the use of surveys (CBS, 2022). The dataset largely consists of characters, with the exception of the Bevolking_1 column, which gives a numerical value for each row representing the amount of people that had the same results. Age, which can be a numerical value, has become an ordinal character as ages are grouped for every ten years. Education level is ordinal as well, but all other characters remain nominal.

The other two datasets, reg_mig_dataset and reg_edu_dataset, are very similar in format and variables. Both contain regional information for each of their variables, but they differ in the type of information. The first dataset contains demographic information, numerical values representing which percentage of the population has which background for each province. The second dataset contains SES information, also numerical values representing percentages, only now they represent the distribution of the population over different education levels for each province. The dataset on education has a column for each level of education. Both datasets have been sourced with open source data and/or administrative sources (CBS, 2025; AlleCijfers, 2026).

Part 3 - Quantifying

3.1 Data cleaning

Renaming unclear characters in edu_mig_dataset

The size of the dataset on education level and different migration backgrounds was so significant that all character values were replaced with codes. In order to have a meaningful dataset, all these codes were replaced with the character value the cells previously contained.

Eliminating unnecessary observations and variables in edu_mig_dataset

Furthermore, the same dataset contained various variables and observations that were irrelevant to the analysis, like migration background data for singular countries and a column for sex, which only contained "total women and men" for all observations. By removing these unnecessary elements, the dataset is easier to use and has become one-fourth its original size.

Renaming inconsistent characters in reg_edu_dataset

The dataset containing data on education levels for each province is to be used to compare regional education levels with the regional migration levels (reg_mig_dataset). However, these two datasets have used different names to address the same provinces, which can cause complications. Therefore, the names in the reg_edu_dataset have been replaced with those of their counterparts in the reg_mig_dataset to ensure uniformity.

Eliminating unnecessary years in reg_edu_dataset

The second change made to this dataset is the elimination of observations corresponding to the years 2023 and 2025. 2023 was eliminated since this dataset, as previously mentioned, will later be used for comparison and the dataset it will be compared to only contains the data for one year. 2025 was eliminated because it contained solely NA's, which made comparison impossible.

Eliminating unnecessary variables in reg_mig_dataset

The dataset on the proportion of people with a migration background per province contained numerous variables that were irrelevant for the intended use of the dataset. For example, the proportion of European migrants and a distinction between first and second generation migrants. These were meaningful data, however, they did not fit the purpose of this dataset.

3.2 Generate necessary variables

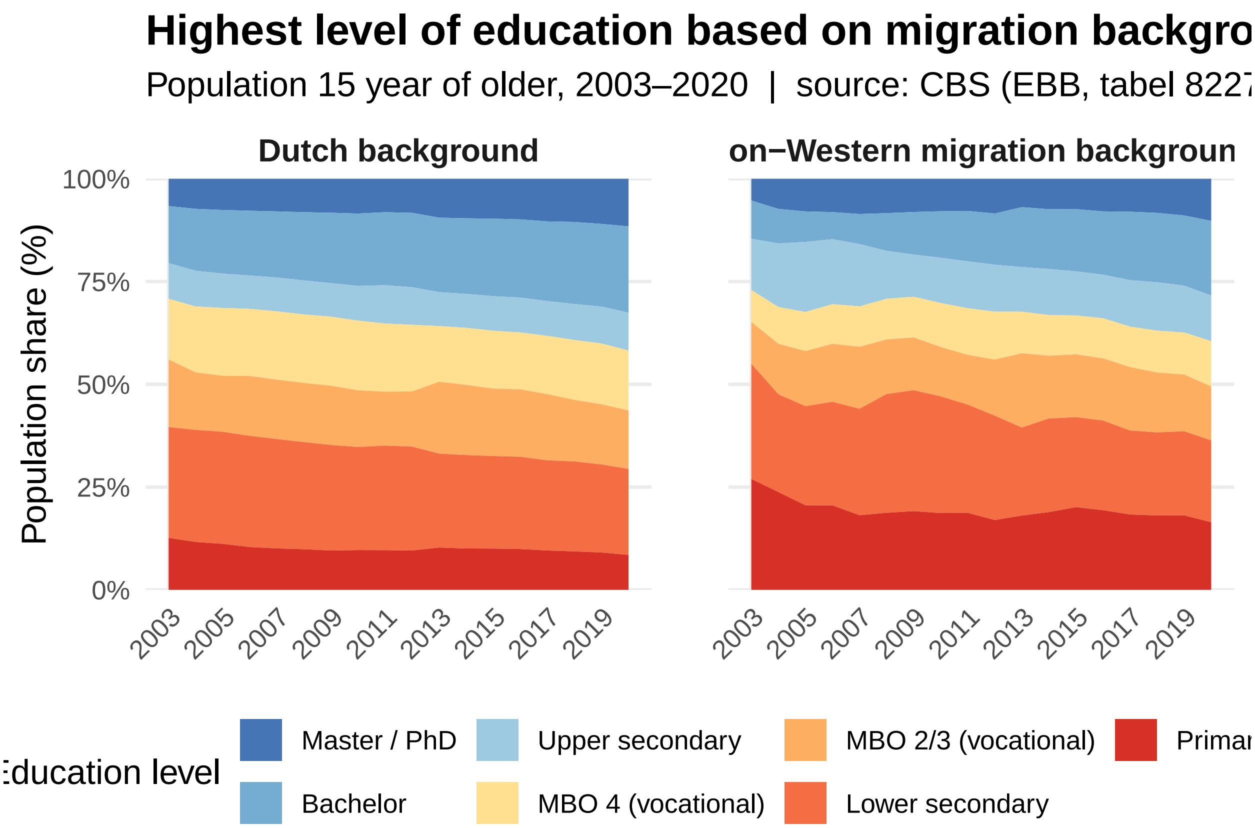
Variable 1: Percentage of the total of Dutch people and non-Western migrants in each level of education

This variable was taken from the cleaned version of the edu_mig_dataset. For the purposes of comparison, this variable was created to demonstrate how Dutch people and non-Western migrants are distributed across different levels of education and how this changes over time. In order to avoid bias because of group size, since there are more Dutch people than migrants, the total of Dutch people for each year was calculated and then the percentage proportion of Dutch people in each education level was calculated. The same was done for non-Western migrants. This way it became clear how each group is distributed across the different levels of education. This resulted in the dataset new_var_2.

Variable 2: Percentage of people that have obtained a hbo- or wo-bachelor for different age groups in 2020

The second variable has also been created with data from the cleaned edu_mig_dataset. This variable was created with the purpose of visualizing sub-population variation, specifically the difference between age groups. The most recent year has been picked to create this variable, which was 2020. A hbo- or wo-bachelor has been selected because it qualifies as a high education. The data was grouped by age and migration background, excluding 15 to 25 year olds, since they would give a distorted image, as a part of this group is too young to obtain a this level of education. Then a total of people per age group was computed, which was then used to calculate the percentage of their age group that obtained a hbo- or wo-bachelor. This resulted in a dataset called combined_var.

3.3 Visualize temporal variation



In the two plots above is shown how the education level of the groups have changed over time. The left plot shows how the education level of the people with a Dutch background changed over time, and the right plot shows the same for people with a non-Western migration background. For the creation of these plots the first variable created earlier was used.

The plot for the people with a Dutch background shows that over time a bigger share of the population obtained a degree from a university or HBO. This is for mostly the bachelor degrees, but also for the master- and doctor degrees there is an increase of the population share. The biggest decrease is seen for the group of people that have a lower secondary education or Vmbo-g/t.

The plot for the people with a non-Western migration background shows that over time a much bigger share of the population got a higher degree than in previous years. The biggest increase is seen in the group of people that have a bachelor degree. There is also a big decrease in the group of people that only finished primary education.

These plots show that the education level of people with a non-Western migration background is becoming closer to the the education level of people with a Dutch background.

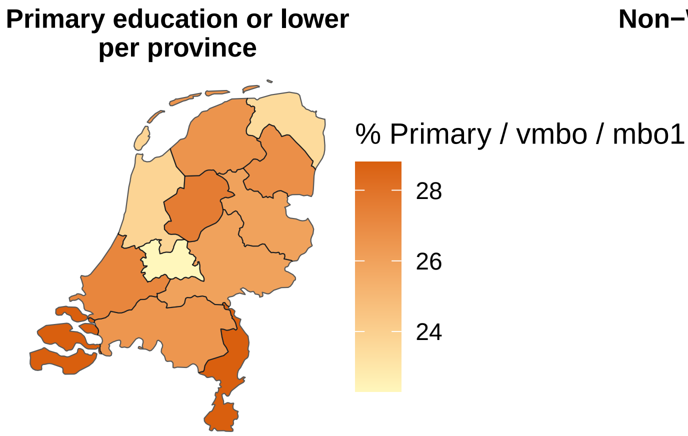
3.4 Visualize spatial variation

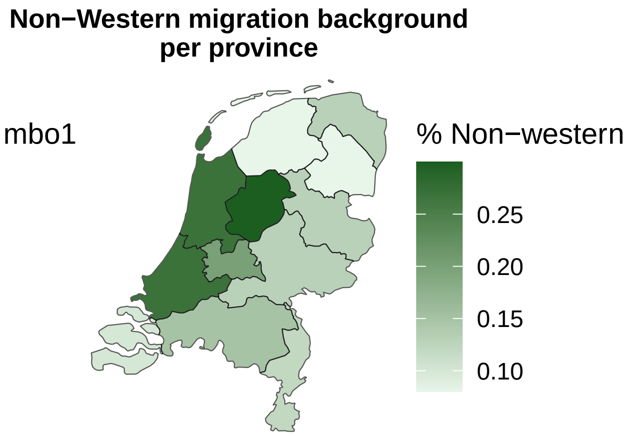
```
# Kolommamen opslaan
basis_col <- "Onderwijs|Naar woonregio|Hoogstbehaald onderwijsniveau|Basisonderwijs, vmbo, mbo1 (%)"

# Kaartdata
```

```r
nl_provinces <- ne_states(country = "Netherlands", returnclass = "sf")
nl_provinces <- nl_provinces[!nl_provinces$name %in% c("St. Eustatius", "Saba"), ]

# Crop
nl_provinces <- st_crop(nl_provinces, xmin = 3.2, xmax = 7.3, ymin = 50.7, ymax = 53.6)
```

```
## Warning: attribute variables are assumed to be spatially constant throughout
## all geometries
```

```r
# Herstel namen naar Nederlands
nl_provinces$name <- recode(nl_provinces$name,
  "North Brabant" = "Noord-Brabant",
  "South Holland" = "Zuid-Holland",
  "North Holland" = "Noord-Holland"
)

# Joins
spatial_var_edu <- right_join(nl_provinces, reg_edu_dataset_cl, by = c("name" = "Regio's"))
spatial_var_mig <- right_join(nl_provinces, reg_mig_dataset_cl, by = c("name" = "Provincienaam"))

# Plot 1
plot_edu <- ggplot(spatial_var_edu) +
  geom_sf(aes(fill = .data[[basis_col]])) +
  scale_fill_gradient(low = "#fff7bc", high = "#d95f0e") +
  labs(
    title = "Primary education or lower\nper province",
    fill  = "% Primary / vmbo / mbo1"
  ) +
  theme_void() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 10))

# Plot 2
plot_mig <- ggplot(spatial_var_mig) +
  geom_sf(aes(fill = `% Herkomst buiten Europa`)) +
  scale_fill_gradient(low = "#e8f5e9", high = "#1b5e20") +
  labs(
    title = "Non-Western migration background\nper province",
    fill  = "% Non-western"
  ) +
  theme_void() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 10))

# Combineren

combined <- plot_edu | plot_mig
combined
```

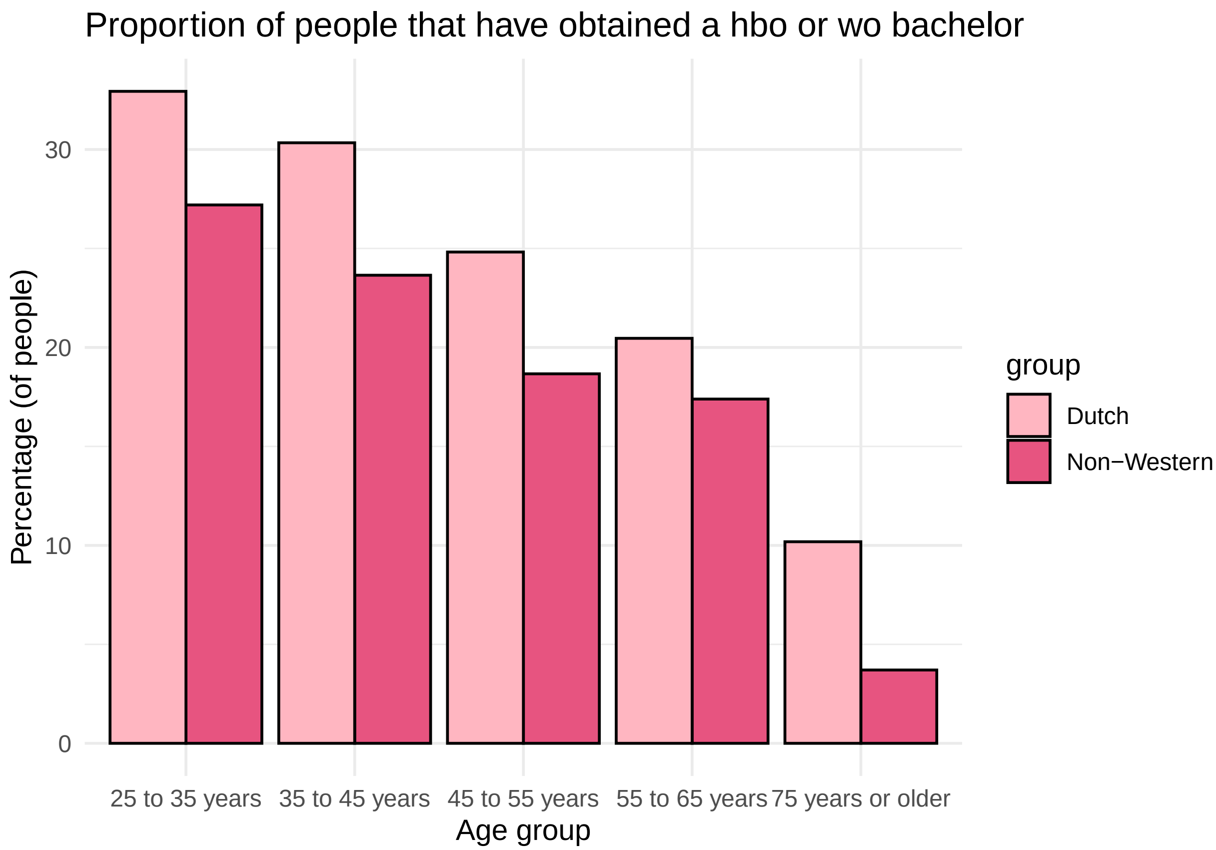




The left map shows the percentage of the population to have either only received primary education, vmbo-g/t or lower secondary education. The map shows that specifically in North-Holland, Utrecht and Groningen the education level is higher than in other parts of the country.

The map on the right shows the percentage of the population who have a non-Western migration background. The map shows that the percentage is the highest in the provinces that are a part of the Randstad. This shows that migration is mostly centered around these parts of the country. The biggest cities of the country are also in the provinces that have the highest percentage.

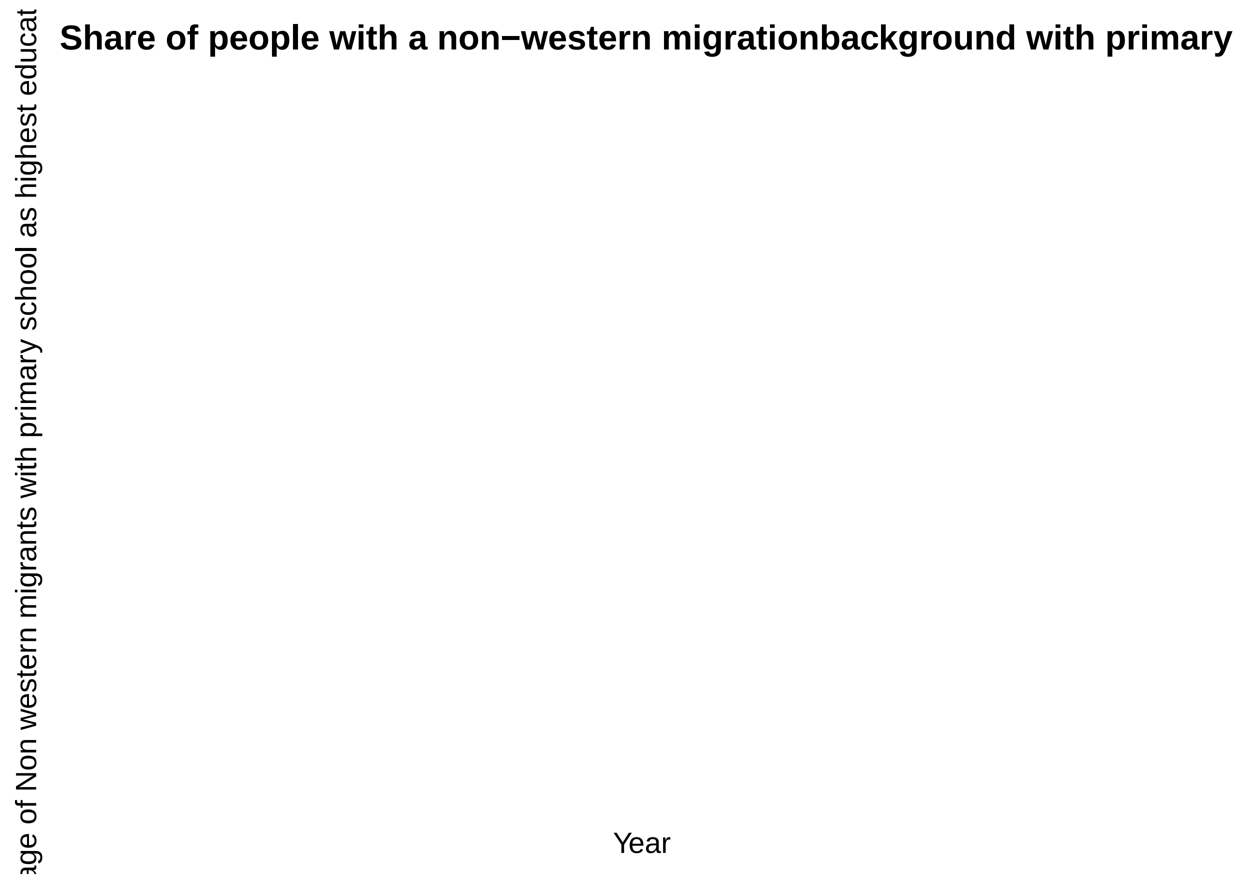
3.5 Visualize sub-population variation



The bar plot pictured above visualizes the percentage of people, both Dutch and with a migration background, that have obtained a hbo- or wo-bachelor in 2020 per age group. This is relevant to the social problem analysis, as it demonstrates not only the difference between Dutch people and non-Western migrants in their participation in higher education, but also whether this differs between different ages. It is clear that, in general, the percentage of people hbo- or wo-bachelor has a negative correlation with age, demonstrating that the younger generations are becoming increasingly high educated. However, the proportional difference between Dutch people and non-Western migrants seems to stay approximately seven percent, except for the 55 to 65 years group. These findings show that the gap does not seem to be closing in the younger generations.

3.6 Event analysis

```
## Warning in max(event_data$pct): no non-missing arguments to max; returning -Inf
```



Find some kind of event in order to show changes caused by this event: Analyze the relationship between two variables. will be a line graph, not sure which dataset yet. (text needs to be removed) Here you provide a description of why the plot above is relevant to your specific social problem. (text needs to be removed)

Part 4 - Discussion

4.1 Discuss your findings

Our analysis shows that people with non-Western migration backgrounds in the Netherlands have lower educational levels than those with Western or Dutch backgrounds. Our data shows that non-Western migrants make up a much larger share of people with only basic education, while university degrees (HBO/WO) are much more common among Dutch-born people. We can also conclude that the age gap in education is the largest among older age groups (75+ years: 10% Dutch vs. 3% non-Western), but becomes smaller for younger generations (25-35 years: 32% Dutch vs. 27% non Western). This shows that younger people with non-Western backgrounds are getting more university degrees than older generations did. The trend for non-Western migrants with only primary education as the highest achieved level, shows an initial decline in 2006 after the introduction of the Wet inburgering policy. The trend then levels off at around 18%-20%, which means that this policy only helped a little. Our map with geographic differences across provinces shows variation in where non-Western migrants live. But the educational gap exists in all regions, even where non-Western populations are smaller. This means that location alone does not explain the educational inequality. For policy makers this suggests that investments in Dutch language lessons/tutoring programs in primary and secondary school may reduce these educational gaps. Because Dutch language skills are essential for success in secondary school and university exams.

Part 5

Reproducibility make sure that if someone were to reproduce our work, a change in packages would not matter as the information of which packages we used when is clear. (text needs to be removed)

5.1 Github repository link

Provide the link to your PUBLIC repository here: ... (text needs to be removed) https://github.com/RikdeBruijn/PFE2026

5.2 Reference list

AlleCijfers (2026, June 8). “Ranglijst: autochtoon en migratieachtergrond van de bewoners per provincie in Nederland”. Retrieved June 8, 2026, from: https://allecijfers.nl/ranglijst/autochtoon-en-migratieachtergrond-per-provincie-in-nederland/#google_vignette

CBS (2025, November 7). “Bevolking; hoogstbehaald onderwijsniveau en regio”. Retrieved June 8, 2026, from: https://opendata.cbs.nl/#/CBS/nl/dataset/85525NED/table

CBS (2021, November 16). “Bevolking; onderwijsniveau en -richting 2003-2021”. Retrieved June 10, 2026, from: https://opendata.cbs.nl/#/CBS/nl/dataset/82275NED/table?ts=1782126231056&fromstatweb=true

Wet van 30 november 2006, houdende regels inzake inburgering in de Nederlandse samenleving (Wet inburgering). (2006). In Staatsblad Van Het Koninkrijk Der Nederlanden (No. 625). Staatsblad van het Koninkrijk der Nederlanden. <https://zoek.officielebekendmakingen.nl/stb-2006-625.pdf